Issue

Brazilian culture includes the concept of *jeitinho*, a resourceful and improvisational approach to problem solving that has been associated with creativity (Da Matta, 2001; Ferreira et al., 2012; Nishioka & Akol, 2019). This raises an important question: how do Brazilian students perform on a standardized measure of creative thinking, and what factors are associated with differences in performance?

PISA 2022 introduced the first international assessment of creative thinking, a competency increasingly considered important for learning and problem solving (OECD, 2024). In Brazil, a large and socioeconomically diverse education system, understanding whether socioeconomic and student characteristics are associated with creative thinking performance can help identify disparities and inform educational support.

This analysis addresses two questions:

- **RQ1 (Prediction):** To what extent can socioeconomic status, math and reading achievement, curiosity, belonging, and teacher support predict high creative thinking performance?
- **RQ2 (Clustering):** Do student profiles based on these characteristics differ in creative thinking outcomes?

Key Findings

- **Sample:** 7,728 Brazilian 15-year-olds were retained after merging the PISA 2022 Student Questionnaire and Creative Thinking files and removing records with missing core variables (from an initial merged sample of 10,798).
- **Creative thinking performance increased with socioeconomic status.** About 9% of students in the lowest SES group were high performers, compared with about 48% in the highest SES group.
- **Four student profiles were identified through clustering, with large differences in creative thinking.** About 61% of students in the most advantaged profile were high performers, compared with about 8% in the most disadvantaged profiles. These differences were statistically significant.

- **Higher SES, math and reading achievement, and curiosity were positively associated with higher creative thinking performance.** Teacher support showed an unexpected negative relationship, while sense of belonging was not statistically significant.
- **Both prediction models performed well.** Logistic regression reached 83.5% accuracy (AUC = .898), while random forest reached 83.6% accuracy (AUC = .888).
- **The models were better at identifying students who were not high performers than identifying high performers.** They identified about 55–58% of high performers compared to 92–93% of other students.

Actionable recommendations

- **Prioritize equitable access to creative-thinking opportunities in lower-SES schools**, including open-ended activities, project-based learning, and arts integration, given the large socioeconomic differences observed in creative-thinking performance.
- **Use SES, academic achievement, and curiosity as broad indicators of where additional support may be needed.** Model predictions should guide general support rather than individual high-stakes decisions because the models did not identify every high performer.
- **Provide opportunities that encourage curiosity and exploration**, since curiosity was positively associated with high creative-thinking performance.
- **Further investigate the unexpected negative relationship with teacher support** before making recommendations based on this finding. The analysis shows a negative association, not that teacher support causes lower creative-thinking performance.

Appendices

Appendix A: Analysis

Introduction

Creativity and resourcefulness have a distinct place in Brazilian cultural approaches to everyday problem solving, reflected in the concept of *jeitinho brasileiro*. *Jeitinho brasileiro* is a culturally recognized approach to navigating problems and constraints that can involve creative, flexible, and unconventional solutions (Ferreira et al., 2012). Research with Brazilian international students has similarly connected its creative dimension with resourcefulness and finding new ways to solve problems (Nishioka & Akol, 2019). These perspectives highlight the relevance of creative thinking in the Brazilian cultural context and its connection to everyday problem solving.

Creative thinking is increasingly recognized as an important educational outcome because students need to generate ideas, adapt to unfamiliar problems, and evaluate possible solutions in school and beyond. PISA 2022 assessed creative thinking internationally for the first time, defining it as students' capacity to generate diverse and original ideas and to evaluate and improve those ideas. The assessment included open-ended tasks in written expression, visual expression, social problem solving, and scientific problem solving (OECD, 2024). This expanded PISA beyond its traditional focus on mathematics, reading, and science and created a new opportunity to examine how students' backgrounds, academic achievement, and attitudes relate to creative thinking performance.

Brazil provides an interesting setting for examining differences in creative thinking because socioeconomic inequalities are closely related to students' educational opportunities and academic outcomes. Previous research has documented substantial differences in educational achievement by socioeconomic background in Brazil, including unequal access to educational resources such as experienced and qualified teachers (Guilherme et al., 2024). Similar differences appear in creative thinking. In PISA 2022, socioeconomically advantaged Brazilian students scored 11.4 points higher in creative thinking than disadvantaged students, compared with an average gap of 9.5 points across OECD countries (OECD, 2024). Socioeconomic status also accounted for approximately 12% of the variation in creative thinking performance within Brazil.

Rather than focusing only on Brazil's position in international rankings, this project examines differences in creative thinking performance among students within the country. Using Educational Data Mining and Learning Analytics methods, this study examines how socioeconomic status, mathematics and reading achievement, curiosity, sense of belonging, and teacher support relate to high creative thinking performance. It also examines how students with similar socioeconomic and academic characteristics can be grouped into profiles and compares creative thinking outcomes across those groups.

Literature Review

The PISA 2022 creative thinking framework treats creativity as a competence that can be demonstrated through generating, evaluating, and improving ideas rather than as a fixed personality trait (OECD, 2023). This approach is relevant to education because it suggests that students can develop and demonstrate creative thinking skills through learning experiences and opportunities. PISA 2022 also found that academic achievement and creative thinking performance are related rather than competing outcomes, meaning students can perform strongly in both traditional academic subjects and creative thinking tasks (OECD, 2024).

Socioeconomic inequality is an important part of the PISA creative thinking findings. Across OECD countries, socioeconomically advantaged students scored about 9.5 points higher in creative thinking than disadvantaged students, although socioeconomic status was generally less strongly related to creative thinking than to mathematics, reading, and science (OECD, 2024). These findings suggest that creative thinking performance is influenced by students' broader educational and socioeconomic contexts. Examining this relationship within Brazil can therefore provide a clearer picture of how socioeconomic differences are associated with creative thinking opportunities among Brazilian students.

Creative thinking performance is also related to academic achievement. PISA 2022 found that students who performed well in mathematics and reading also tended to perform well in creative thinking. (OECD, 2024). For this reason, mathematics, reading, and socioeconomic status are also included in the analysis.

Additionally, the study considers student attitudes and school experiences. PISA has identified relationships between creative thinking and students' beliefs, attitudes, and classroom experiences (OECD, 2024). Based on these findings, curiosity, sense of belonging, and teacher support are included to examine how they relate to creative thinking performance.

Data Source and Variable Verification

Data were sourced from the OECD PISA 2022 Database. The Student Questionnaire file (CY08MSP_STU_QQQ) provided socioeconomic status, curiosity, sense of belonging, teacher support, and plausible values for mathematics and reading. The Creative Thinking file (CY08MSP_CRT_COG) provided ten plausible values for creative thinking performance. Records were filtered to Brazil (CNT =

"BRA") and merged using school and student identifiers (CNTSCHID and CNTSTUID), producing an initial merged sample of 10,798 students.

Before variables were selected for analysis, the official PISA 2022 data dictionary (CY08MSP_CODEBOOK_27thJune24.xlsx) was consulted to verify variable names, labels, source files and the meaning of the measures. This step was important because the PISA public-use files contain many closely related questionnaire items, derived indices, weights, and plausible values. Using the codebook helped confirm that the variables included in the analysis corresponded to the intended constructs before the datasets were merged and modeled.

Table 1. Selected variables used in the analysis

Role	PISA variable(s)	Use in analysis
Country filter	CNT	Filter to Brazil only
Identifiers	CNTSCHID, CNTSTUID	Merge student and creative thinking files
Socioeconomic status	ESCS	Predictor; SES quartiles
Mathematics achievement	PV1MATH–PV10MATH	Predictor; averaged for exploratory analysis
Reading achievement	PV1READ–PV10READ	Predictor; averaged for exploratory analysis
Curiosity	CURIOAGR	Predictor / clustering variable
Sense of belonging	BELONG	Predictor / clustering variable
Teacher support	TEACHSUP	Predictor / clustering variable
Creative thinking	PV1CRTH_NC–PV10CRTH_NC	Outcome; averaged for exploratory analysis

Data Preparation

PISA provides multiple possible scores for mathematics, reading, and creative thinking rather than one single score for each student. For this project, these scores were averaged to create one score for each subject. This simplified the analysis, although a full PISA analysis would examine the scores separately and combine the results using PISA's recommended procedures.

Students with missing data on any of the main variables were removed, reducing the sample by 3,070 students and leaving 7,728 students for the final analysis. Extreme values in the continuous variables were capped at the 1st and 99th percentiles to reduce the influence of unusually high or low values while keeping all observations in the analysis. For the clustering analysis, variables were also standardized so that differences in their original measurement scales would not affect how the student groups were formed.

For descriptive comparisons, socioeconomic status was divided into four groups, from the lowest to the highest quartile. Students were classified as high creative thinking performers if their score was in the top 25% of the Brazilian sample, corresponding to a score of 33.84 or higher. This cutoff was created for this analysis and should not be interpreted as an official OECD proficiency level.

RQ1: Predicting High Creative Thinking Performance

To answer RQ1, logistic regression and random forest models were used to predict whether a student was classified as a high creative thinking performer. Both models used the same six predictors: socioeconomic status, mathematics achievement, reading achievement, curiosity, sense of belonging, and teacher support.

The analytic data were divided into a 70% training set and a 30% test set using stratified sampling so that the proportion of high performers was similar in both sets. Logistic regression was used as an interpretable baseline, while a random forest with 500 trees was used as a nonlinear comparison model. Model performance was evaluated on the held-out test set. Five-fold cross-validation was also used as a robustness check.

Both models showed strong discrimination and nearly identical accuracy. Logistic regression achieved 83.5% accuracy and an AUC of .898, while random forest achieved 83.6% accuracy and an AUC of .888. Five-fold cross-validation produced similar AUC values (.894 for logistic regression and .886 for random forest), suggesting that performance was not driven by a single train/test split.

Because only about 25% of students were classified as high performers, accuracy alone can overstate practical performance. Recall was 55.1% for logistic regression and 58.2% for random forest, meaning the models identified only a little more than half of all true high performers. Specificity was much higher at approximately 92–93%. In simple terms, the models were better at identifying students who were not high performers than at finding every high-performing student.

Table 2. Model performance predicting high creative thinking performance

Model	Accuracy	Recall	F1	AUC
Logistic Regression	0.835	0.551	0.625	0.898
Random Forest	0.836	0.582	0.639	0.888

Logistic regression showed that socioeconomic status, mathematics achievement, reading achievement, and curiosity were significantly and positively associated with high creative thinking performance. Sense of belonging was not statistically significant, while teacher support showed a small but statistically significant negative association (See Table 3). Random forest variable importance similarly emphasized mathematics achievement, reading achievement, and socioeconomic status. The two models point to achievement and socioeconomic background as the strongest overall signals, with curiosity adding additional information

Table 3. Logistic regression coefficients predicting high creative thinking performance

Predictor	Coefficient (b)	p-value	Direction
ESCS (SES)	0.300	< .001	Positive
Math achievement	0.013	< .001	Positive
Reading achievement	0.011	< .001	Positive
Curiosity	0.172	< .001	Positive
Sense of belonging	-0.033	.470	None
Teacher support	-0.107	.006	Negative

RQ2: Student Profiles and Creative Thinking

To answer RQ2, k-means clustering was used to identify groups of students with similar profiles across the six standardized variables. The outcome variable (creative thinking performance) was not used to create the clusters. Creative thinking outcomes were compared only after the student profiles had been formed.

The number of clusters was evaluated using both the elbow method and silhouette score. The elbow plot showed diminishing improvements beyond approximately four clusters, while the silhouette method favored two clusters. A four-cluster solution was retained because it provided more detailed and educationally interpretable student profiles.

The four profiles differed substantially in creative thinking outcomes. Cluster 1, the most advantaged profile, included 2,234 students and had the highest average creative thinking score. About 61.2% of students in this group were high performers. By contrast, only 7.3% of students in Cluster 2 and 7.7% in Cluster 4 were high performers. Cluster 3 fell between these groups, with 27.9% high performers (see Table 4). A one-way ANOVA confirmed that mean creative thinking scores differed significantly across the four clusters, $F(3, 7724) = 1674$, $p < .001$.

Table 4. Student profiles and creative thinking outcomes

Cluster	n	Mean CT	Mean ESCS	Mean Math	% High CT
1	2,234	36	-0.11	473.9	61.2%
2	2,083	21	-1.42	356.1	7.3%
3	761	27.8	-0.63	402.7	27.9%
4	2,650	20.4	-1.18	351.3	7.7%

Appendix B: Methodology and Interpretation

Analytical Framework

This project followed an EDM/LA knowledge-discovery workflow adapted to the PISA data: data acquisition, variable verification, filtering and merging, data preparation, modeling, interpretation, and visualization (Romero & Ventura, 2020). Variable selection was checked against the PISA 2022 codebook before analysis to ensure that the measures used in the models matched the intended constructs.

Prediction and clustering were selected because they answer different but related questions. Predictive modeling uses student characteristics to distinguish between outcome groups (Brooks &

Thompson, 2022). In this study, logistic regression and random forest were used to classify students as high or not high creative-thinking performers based on socioeconomic status, mathematics and reading achievement, curiosity, sense of belonging, and teacher support. Logistic regression provided interpretable coefficients and significance tests, while random forest allowed for more flexible patterns and measures of variable importance (Breiman, 2001).

Random forest variable importance was evaluated using Mean Decrease Accuracy, which measures how much prediction accuracy declines when a predictor is disrupted, and Mean Decrease Gini, which reflects how much a predictor contributes to separating students into outcome groups within the trees.

Clustering addressed a different question by identifying groups of students with similar characteristics without using creative thinking to form the groups. Amershi and Conati (2009) describe k-means clustering as grouping observations according to their similarity in a normalized feature space. In this study, students were clustered using standardized measures of socioeconomic status, mathematics, reading, curiosity, belonging, and teacher support. Creative thinking outcomes were then compared across the resulting student profiles.

Each method has different strengths and limitations. Logistic regression is relatively easy to interpret but can fail to capture complex or nonlinear relationships. Random forest can model more complex patterns but is less directly interpretable. K-means provides a straightforward way to identify student profiles based on similarity, but its results depend on the selected number of clusters and the distances between observations. Variables were therefore standardized before clustering so that differences in their original measurement scales would not influence the cluster solution. The number of clusters was evaluated using both the elbow method and silhouette scores. Four clusters were retained because they provided more detailed and educationally interpretable profiles, although the silhouette method favored two clusters.

Using both approaches provides complementary evidence. The prediction models identified socioeconomic status and academic achievement as important indicators of high creative thinking performance, while the clustering results showed that profiles with stronger socioeconomic and academic characteristics also contained much higher shares of high creative thinking performers. The methods provide two perspectives on patterns of creative thinking performance within the Brazilian sample.

Interpretation of Findings

Overall, the prediction and clustering analyses point to the same pattern: creative thinking performance is consistently associated with socioeconomic status and academic achievement. Consistent with this pattern, random forest variable importance ranked mathematics, reading, and socioeconomic status as the strongest predictors across the accuracy- and Gini-based importance measures. About 48% of students in the highest SES quartile were classified as high creative thinking performers, compared with about 9% in the lowest quartile. The clustering analysis showed a similar difference. Approximately 61% of students in the most advantaged cluster were high performers, compared with only about 7–8% in the two most disadvantaged clusters. These results suggest substantial socioeconomic and academic differences in creative thinking outcomes within the Brazilian sample.

Curiosity was also positively associated with high creative thinking performance even after socioeconomic status and mathematics and reading achievement were included in the model. This indicates that differences in creative thinking are not associated only with academic performance or socioeconomic background. However, because PISA is cross-sectional, the analysis cannot determine whether curiosity contributes to creative thinking, creative experiences increase curiosity, or both are influenced by other educational experiences.

The prediction results also show an important limitation of using these models for individual decisions. Although both models had strong overall discrimination, they identified only about 55–58% of high performers, while correctly identifying more than 90% of students who were not high performers. The models are therefore more useful for identifying broad patterns in the sample than for screening individual students. Using them for high-stakes decisions could overlook many students with strong creative thinking performance.

Finally, the clustering results show that socioeconomic status, academic achievement, and student characteristics do not operate in isolation. Instead, they combine into distinct student profiles associated with substantially different creative thinking outcomes. These differences raise important equity questions about whether students from different socioeconomic backgrounds have similar opportunities to engage in open-ended problem solving, exploration, and idea generation. The findings do not establish that unequal opportunities caused the observed differences, but they identify this as an important area for further investigation.

Interpreting the Teacher-Support Finding

One unexpected result was the negative coefficient for teacher support. The simple correlation between teacher support and creative thinking was close to zero ($r = .03$), but teacher support became negatively associated with high creative thinking performance after socioeconomic status and academic achievement were included in the regression. This pattern could also reflect a statistical suppression effect.

One possible explanation is that students who are struggling academically may receive more teacher support. If so, greater support could appear alongside lower performance even though the support itself is not causing lower creative thinking scores. The PISA teacher-support measure may also capture forms of structured or remedial support that differ from teaching practices specifically intended to encourage originality and creative problem solving. Because PISA is cross-sectional, the direction of the relationship cannot be determined. This finding should therefore be interpreted cautiously and explored in future research.

Limitations

- Cross-sectional design: PISA 2022 is cross-sectional, so the associations identified in this analysis should not be interpreted as causal effects.
- PISA methodology: The analysis did not apply PISA sampling weights or the full plausible-value methodology. Results therefore describe the analytic sample rather than population-representative estimates for all Brazilian 15-year-olds, and uncertainty may be understated.
- Missing data: Complete-case analysis removed 3,070 records, which may introduce selection bias if students with missing or suppressed data differed systematically from those retained.
- Cluster selection: The four-cluster solution was selected for interpretability even though the silhouette score favored two clusters, so other clustering solutions are also reasonable.
- Outcome definition and imbalance: High creative thinking performance was defined as the top 25% of the Brazilian sample rather than an OECD proficiency benchmark. Because only about 25% of students were classified as high performers and the models were not rebalanced, this may have contributed to lower recall for this group.
- Generalizability: The findings are specific to Brazil and should not be generalized to other PISA-participating countries without further testing.

Future Directions

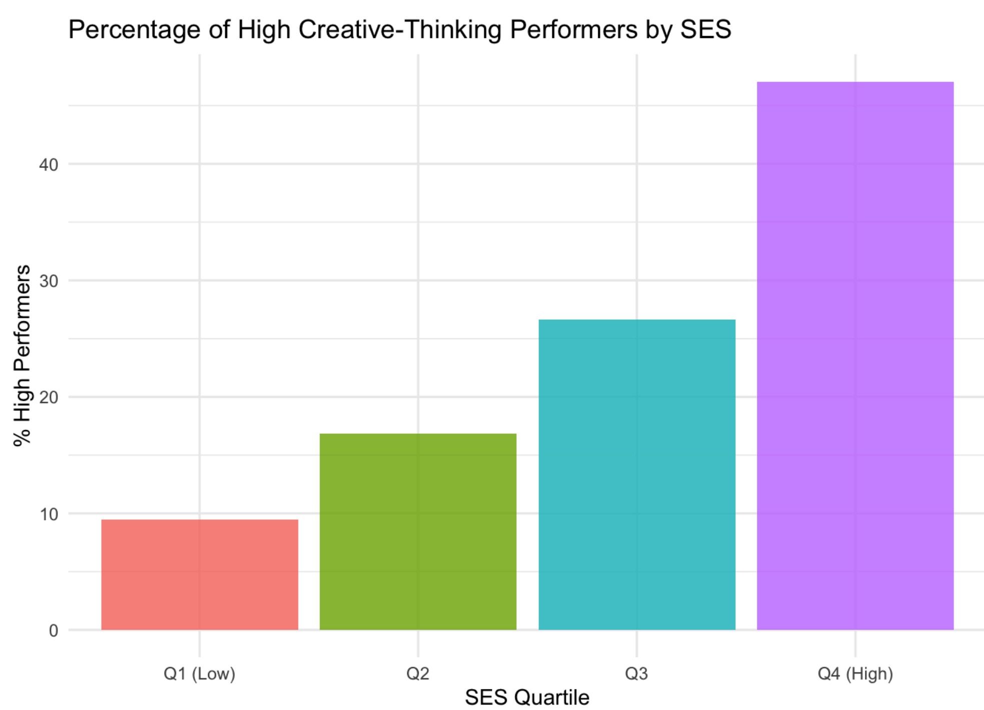
Future research could apply PISA sampling weights and the full plausible-value methodology to produce population-representative estimates. The analysis could also be extended to other countries to determine whether the patterns identified in Brazil are consistent across different contexts. Methodologically, future studies could test class-balancing approaches to improve recall for high performers, compare alternative clustering solutions, and examine interactions such as SES × curiosity. Finally, longitudinal, item-level, or qualitative research could help clarify the unexpected relationship between teacher support and creative thinking performance.

Appendix C: Visuals and Supporting Materials

RStudio knitted HTML outputs for the full analysis are available here: Drive Folder Link
Analysis was conducted with assistance from Claude (Anthropic).

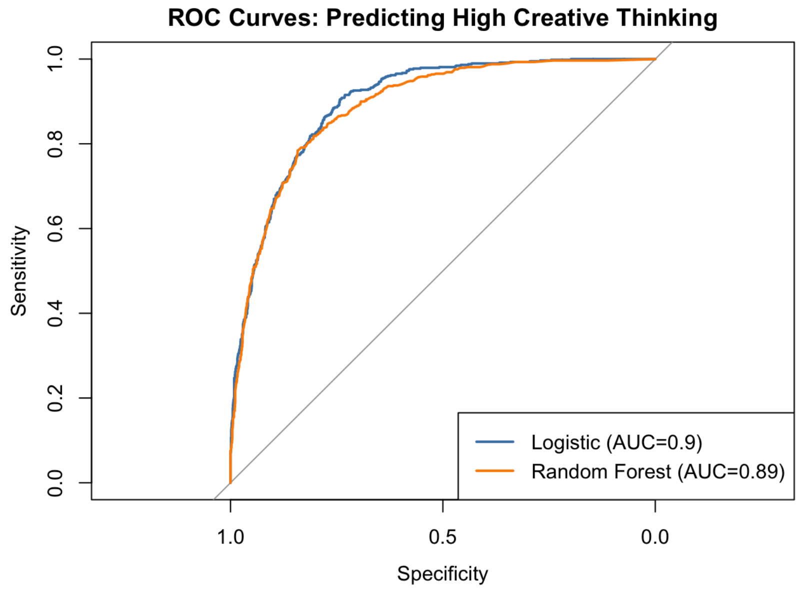
Research Question 1: Prediction (Logistic regression and random forest)

Figure 1. Percentage of high creative thinking performers by SES quartile.



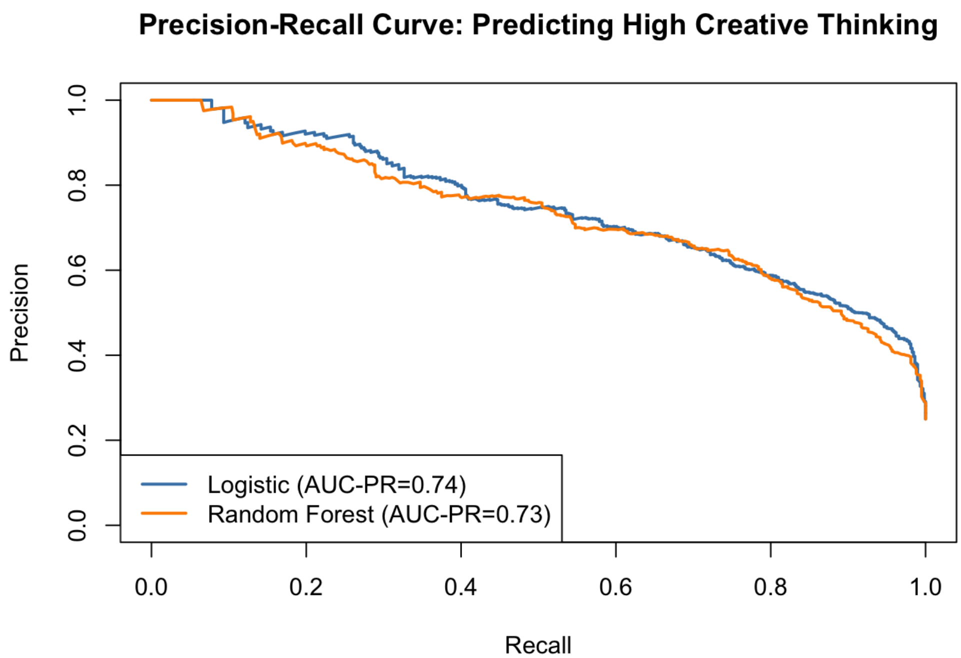
This figure shows the percentage of students classified as high performers within each SES quartile. The large increase from roughly 9% in the lowest quartile to nearly 48% in the highest highlights substantial socioeconomic differences in high creative thinking performance.

Figure 2. ROC curves comparing logistic regression and random forest models.



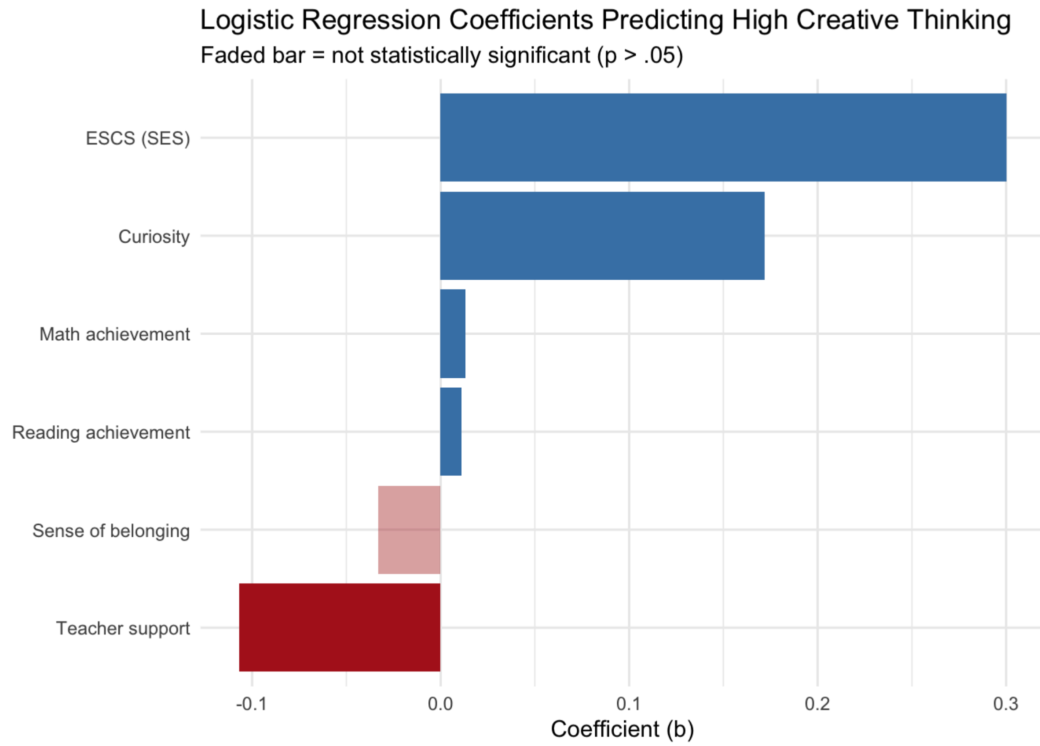
This figure compares the ability of logistic regression and random forest to distinguish high performers from other students across different classification thresholds. The similar and high AUC values show that both models have strong discriminatory performance.

Figure 3. Precision-Recall Curves Comparing Logistic Regression and Random Forest



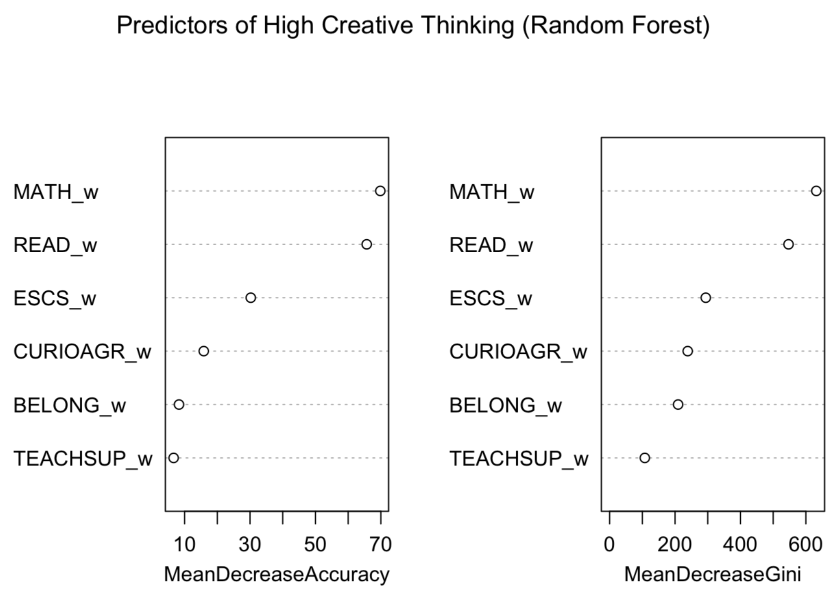
This figure compares model precision and recall across different classification thresholds. It is useful because high performers make up only about 25% of the sample and shows that both models perform similarly when identifying this less common group.

Figure 4. Logistic Regression Coefficients Predicting High Creative-Thinking Performance



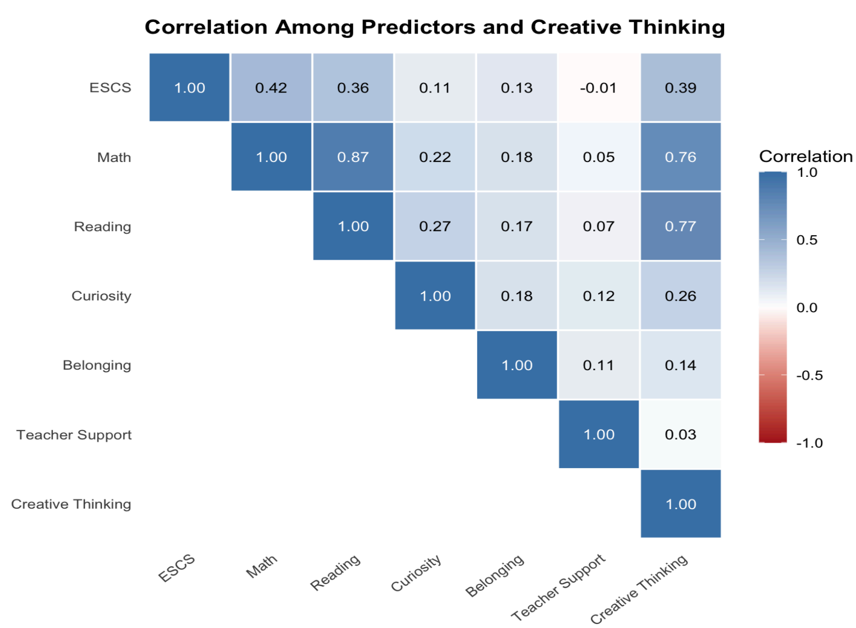
This figure shows the direction of the relationships between each predictor and high creative thinking performance after accounting for the other variables. SES, mathematics, reading, and curiosity have positive associations, while teacher support has a negative association and belonging is not statistically significant.

Figure 5. Random forest variable importance (accuracy-based and Gini-based)



This figure ranks predictors using accuracy- and Gini-based importance measures. Mathematics, reading, and socioeconomic status emerge as the strongest predictors, showing which student characteristics contributed most to the random forest model.

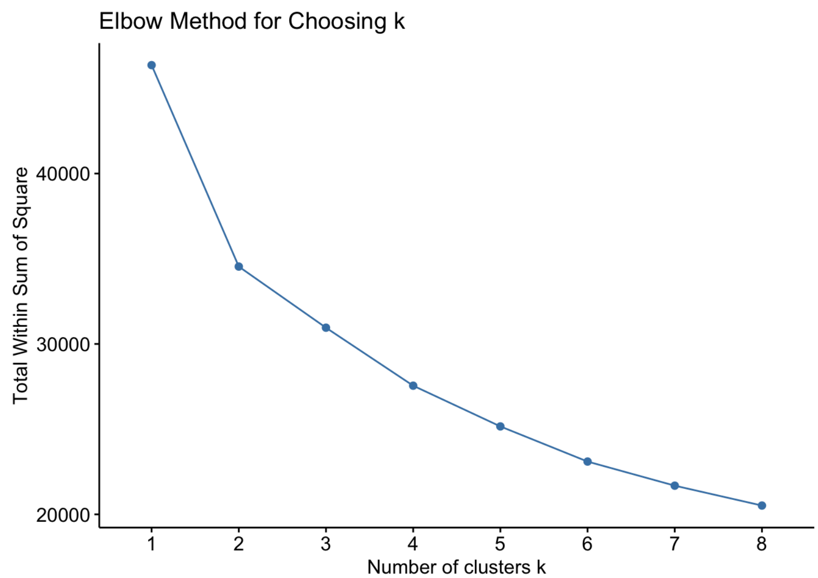
Figure 6. Correlation among predictors and creative thinking.



This figure shows the pairwise relationships among the study variables. Mathematics and reading have the strongest correlations with creative thinking, while teacher support has almost no simple correlation with creative thinking (*r* =.03), providing important context for the regression results.

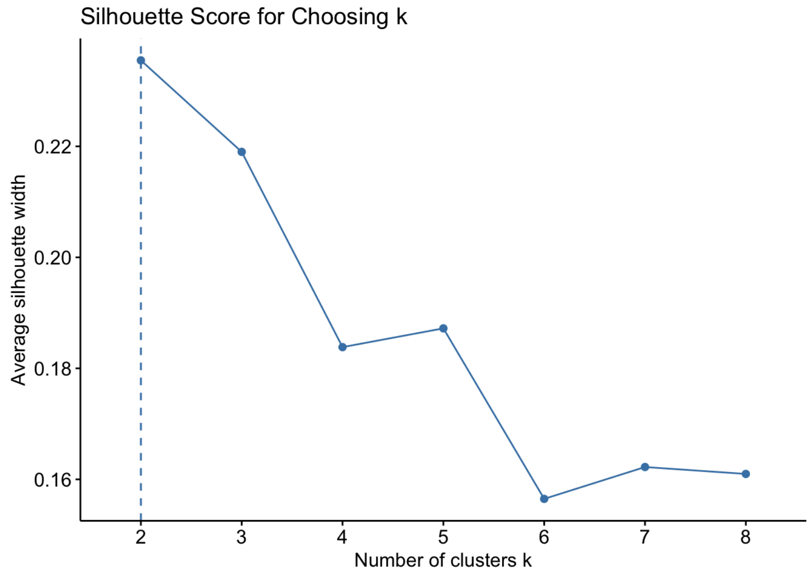
Research Question 2: Clustering

Figure 7. Elbow method for selecting the number of clusters (k)



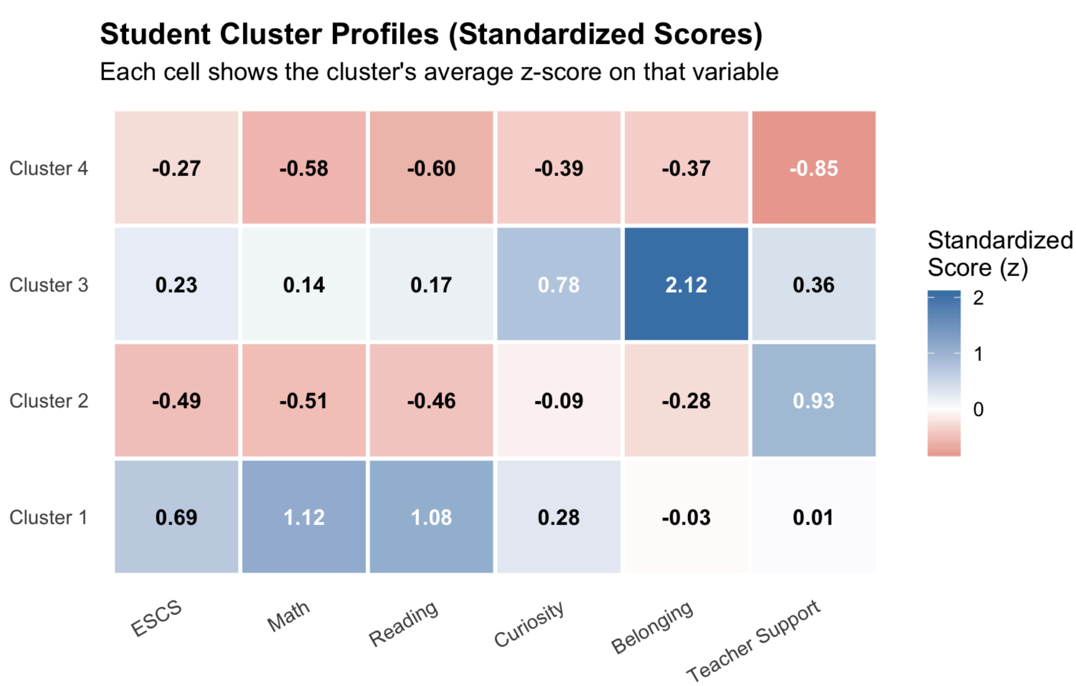
This figure shows how within-cluster variation decreases as the number of clusters increases. The smaller improvements after approximately four clusters helped support the selection of a four-cluster solution.

Figure 8. Silhouette score for selecting the number of clusters (k).



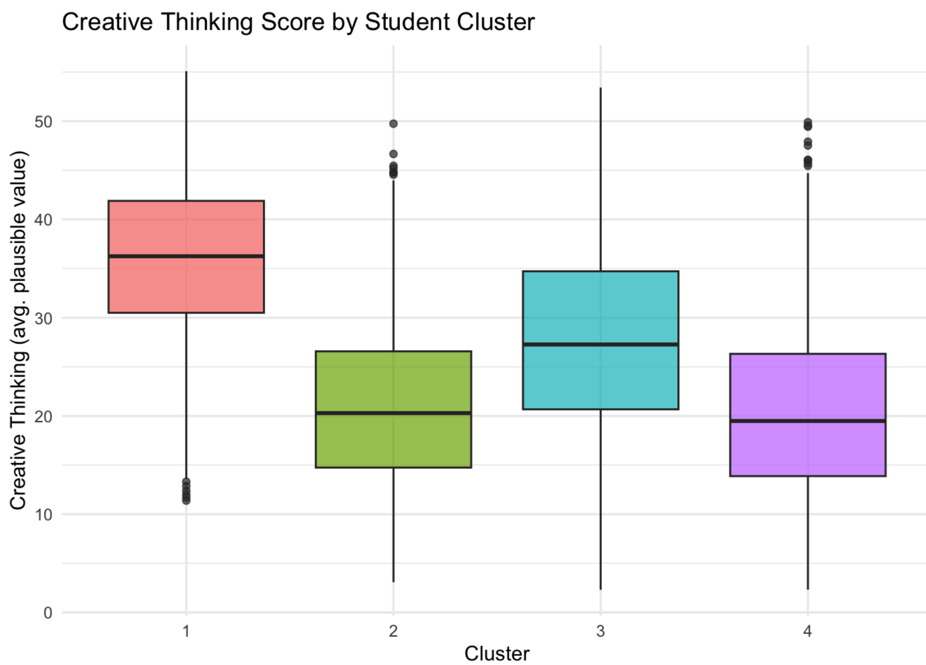
This figure compares how well separated the clusters are for different values of k. Although the highest silhouette score occurs at two clusters, four clusters were retained because they provided more detailed and interpretable student profiles.

Figure 9. Student Cluster Profiles Across Six Variables



This heatmap shows the characteristics that distinguish the four student profiles. Cluster 1 has the highest SES and academic achievement; Cluster 2 has lower SES and achievement but high teacher support; Cluster 3 is distinguished by high curiosity and sense of belonging; and Cluster 4 scores below average across all six characteristics.

Figure 10. Creative thinking score distribution by student cluster.



This figure compares the distribution of creative thinking scores across the four student profiles. Cluster 1 has substantially higher scores than the other clusters, showing that creative thinking outcomes differ across the profiles.

Appendix D: References

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